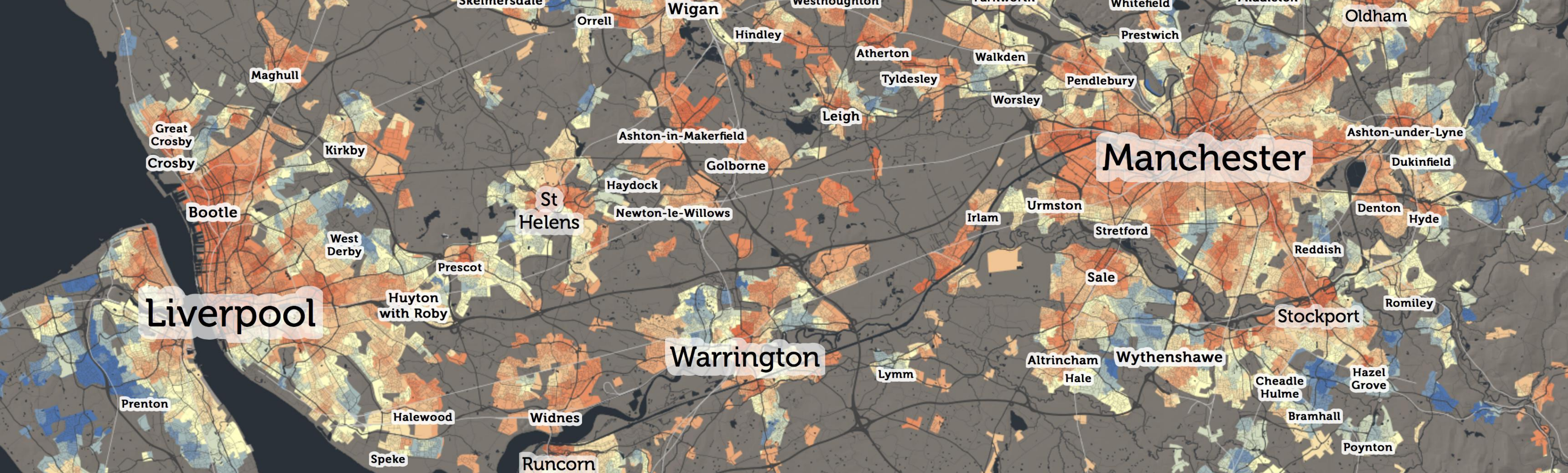




Multiple Linear Regression

ENVS225 Exploring the Social World

Gabriele Filomena

gfilo@liverpool.ac.uk

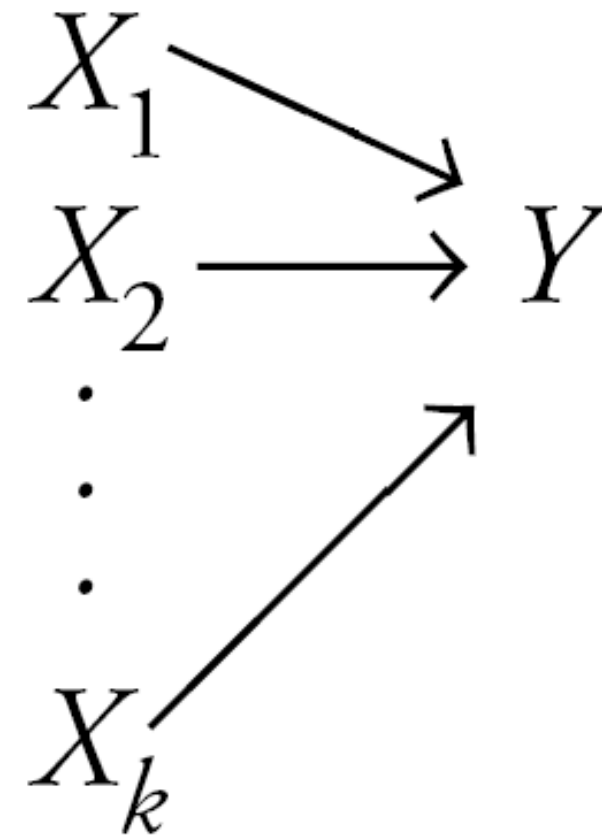
Learning goals

- Learn when to use multiple regression
- Learn how multiple regression extends simple linear regression
- Learn how to use multiple regression in real applications

Simple regression considers the relation between a single explanatory variable and response variable

$$X \rightarrow Y$$

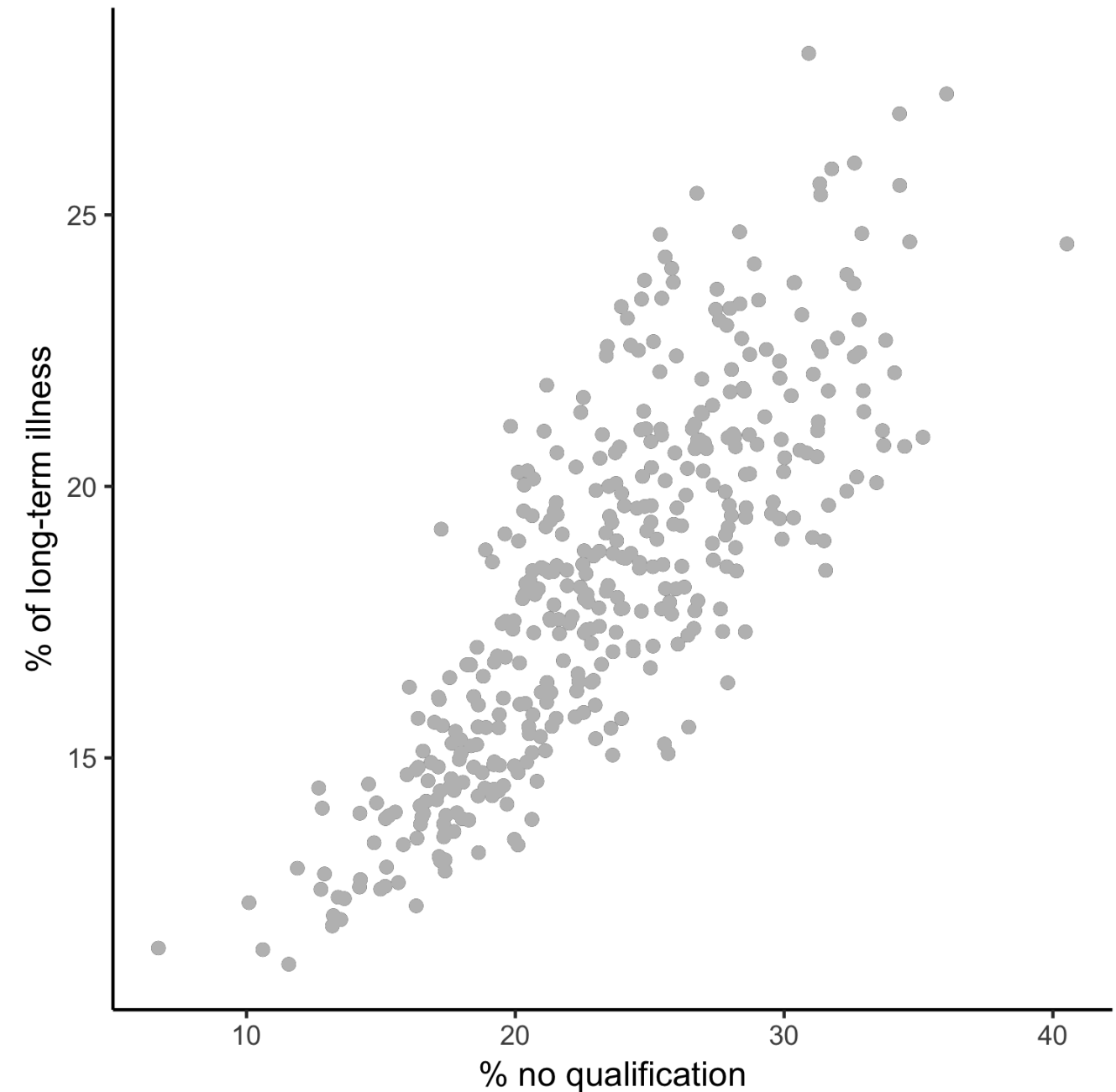
Multiple regression simultaneously considers the relationship between multiple explanatory variables and a variable Y

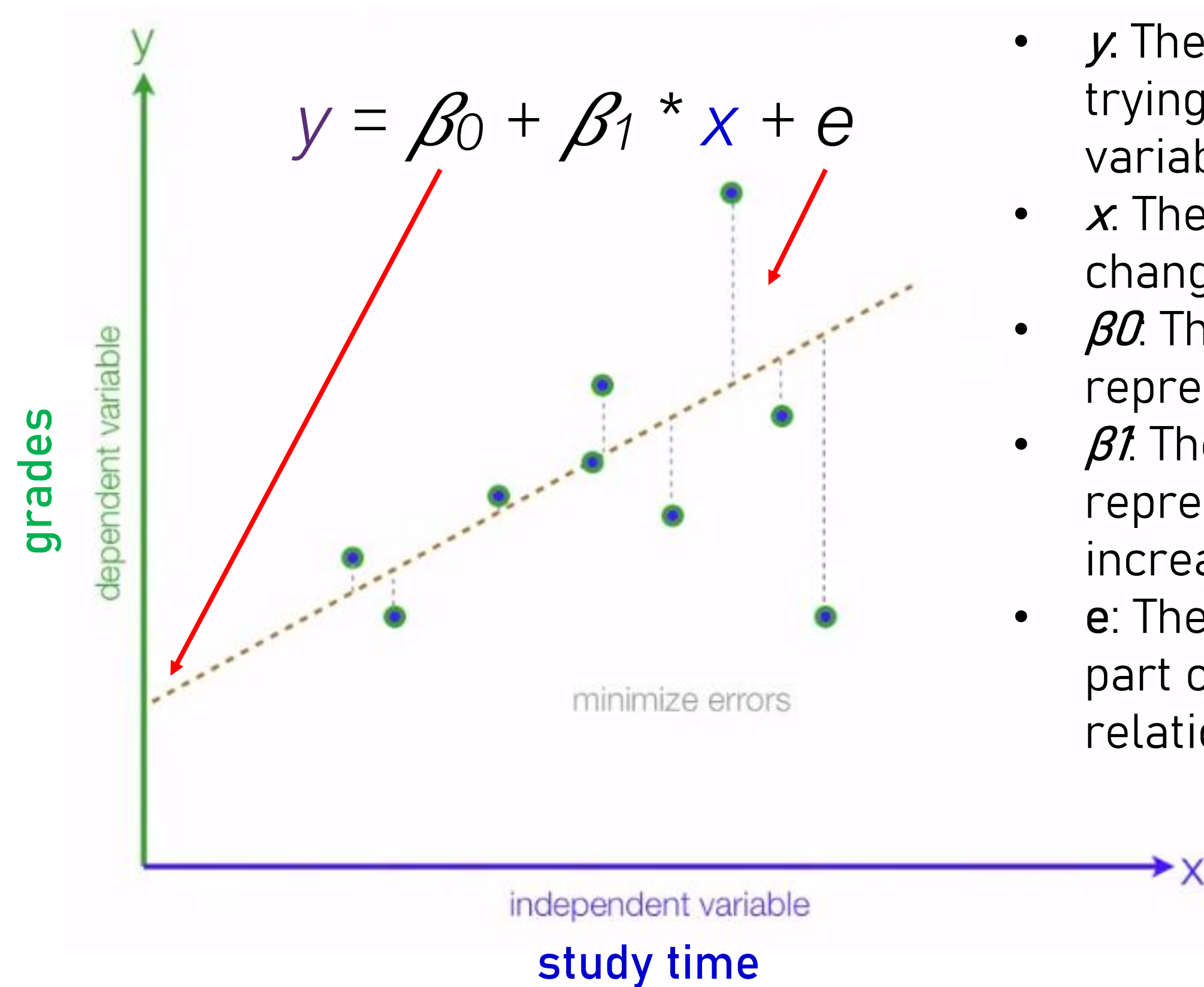


Simple Linear Regression Analysis

It is used to estimate the relationship between **2** continuous variables.

- It measures the relationship between two variables.
- It can predict the value of Y at a certain value of X





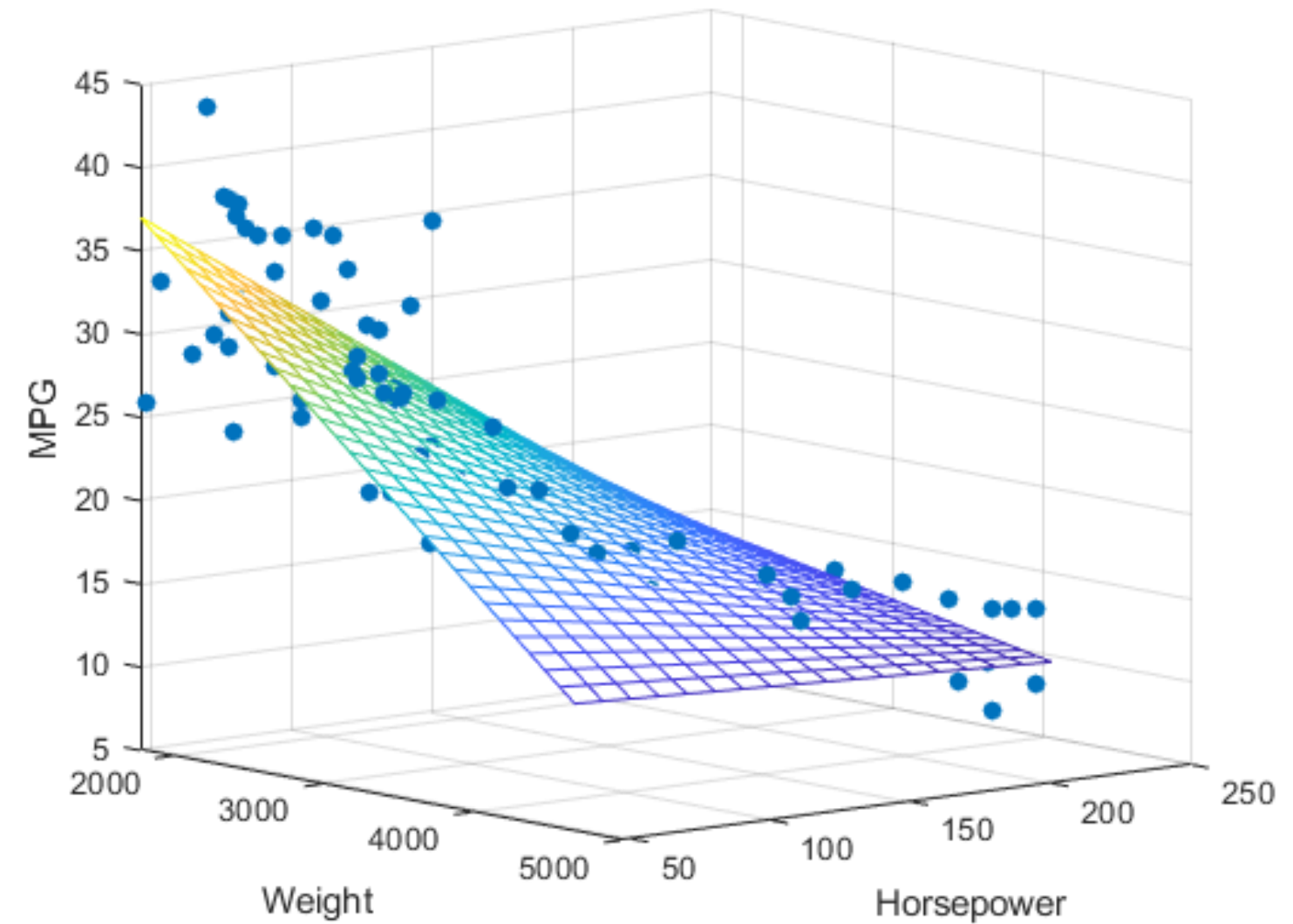
- y : The dependent variable. The outcome we are trying to explain based on the independent variable.
- x : The independent variable used to explain changes in the dependent variable.
- β_0 : The intercept of the regression line. It represents the expected value of y when x is 0.
- β_1 : The slope of the regression line. It represents the change in y for a one-unit increase in x .
- e : The error term (or residual). It represents the part of y that cannot be explained by the linear relationship with x .

Multiple Regression Analysis

Method for studying the relationship between **1 dependent variable** and **2+ independent variables**.

Purposes:

- **Explanation (our focus)**
- Theory building
- Prediction



Simple

vs

Multiple

- One dependent variable Y predicted from one independent variable X .
- One regression coefficient.
- r^2 : proportion of variation in dependent variable Y predictable from X .

- One dependent variable Y predicted from a set of independent variables ($X_1, X_2, \dots, X_k$).
- One regression coefficient for each independent variable.
- R^2 : proportion of variation in Y predictable by set of independent variables (X_s).

Data Requirements

- **One dependent variable -> Continuous**
- Two or more independent variables (explanatory variables).
- Sample size: at least 50, plus 10–20 per independent variable
- Variables must be numerical or categorical converted into **dummy variables** (e.g. religion -> Catholic (y/n -> 1/0), Muslim (y/n -> 1/0 etc.) → **Next week**

Assumptions

- **Independence:** observations (and so their residuals) are independent of one another
- **Normality:** the residuals are normally distributed
- **Homoscedasticity:** the residuals have a constant variance across all values of the independent variables.
- **Linearity:** the relation between the dependent variable and the independent variable is linear when all the other independent variables are held constant.

The Model

The diagram illustrates the components of a multiple regression model equation, $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$. The equation is centered within a light blue rectangular box. Arrows point from descriptive labels to specific parts of the equation: 'Dependent variable' points to Y ; 'Intercept' points to β_0 ; 'Slope coefficient 1' points to β_1 ; 'Independent variable 1' points to X_1 ; 'Slope coefficient 2' points to β_2 ; 'Independent variable 2' points to X_2 ; 'Slope coefficient 3' points to β_3 ; 'Independent variable 3' points to X_3 ; and 'Error' points to ϵ .

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

Labels and their corresponding parts in the equation:

- Dependent variable: Y
- Intercept: β_0
- Slope coefficient 1: β_1
- Independent variable 1: X_1
- Slope coefficient 2: β_2
- Independent variable 2: X_2
- Slope coefficient 3: β_3
- Independent variable 3: X_3
- Error: ϵ

Interpretation

Intercept (β_0)

- The estimated average value of Y when the value of the X s is zero.

$$Y = \beta_0 + \cancel{\beta_1 X 1}^0 + \cancel{\beta_2 X 2}^0 + \cancel{\beta_3 X 3}^0 + \epsilon$$

Slope

- The estimated average change in Y for a one-unit change in X , when all other explanatory variables are held constant

$$Y = \beta_0 + \beta_1 X 1 + \beta_2 X 2 + \beta_3 X 3 + \epsilon$$

Example

$$Y = \beta_0 + \beta_1 X1 + \beta_2 X2 + \beta_3 X3 + \epsilon$$

% of very bad health

% of males

% no
qualification

% higher
professionals

We are trying to find the β s

Output

Call:

```
lm(formula = pct_Very_bad_health ~ pct_No_qualifications + pct_Males +  
    pct_Higher_manager_prof, data = census)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.4903	-0.1369	-0.0352	0.0983	0.7658

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.01799	0.88004	4.57	0.0000071 ***
pct_No_qualifications	0.05296	0.00591	8.96	< 0.00000000000000002 ***
pct_Males	-0.07392	0.01785	-4.14	0.0000440 ***
pct_Higher_manager_prof	-0.01309	0.00494	-2.65	0.0084 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.213 on 327 degrees of freedom

Multiple R-squared: 0.61, Adjusted R-squared: 0.607

F-statistic: 171 on 3 and 327 DF, p-value: <0.00000000000000002

$$Y = \beta_0 + \beta_1 X1 + \beta_2 X2 + \beta_3 X3 + \epsilon$$

% very bad health =

4.01 - 0.074*(% males) + 0.053*(%no qualification) - 0.013*(% higher professional)

Interpretation (other variables held constant)

- β_1 -> A 1%-point increase in the **percentage of males** is associated with a 0.074 percentage-point decrease in the % of population in very bad health
- β_2 -> A 1%-point increase in the **percentage of no qualification** population is associated with a 0.053 percentage-point increase in the % of population in very bad health
- β_0 -> If there were no local percentage of male, no qualification and higher professional population, the percentage of population in very bad health would be 4.01% (very unlikely scenario).

Units of measurement of the dependent and independent variables are important for interpretation purposes!

Hypothesis testing and Significance

For each variable X_i , the null hypothesis:

- H_0 : There is no association between X_i and Y .

vs the alternative hypothesis:

- H_1 : There is an association between X_i and Y .

If the null hypothesis is rejected, there is evidence that there is a significant relationship between X_i and Y .

t-test

- The t-test is performed as a hypothesis test to assess the significance of individual coefficients in the linear regression model.
- We look at the **p-value** of each coefficient β_i .
- If the p-value is less than 0.05, we reject the null hypothesis, otherwise, we do not.

What results should be reported?

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.01799	0.88004	4.57	0.0000071	***
pct_No_qualifications	0.05296	0.00591	8.96	< 0.00000000000000002	***
pct_Males	-0.07392	0.01785	-4.14	0.0000440	***
pct_Higher_manager_prof	-0.01309	0.00494	-2.65	0.0084	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.213 on 327 degrees of freedom

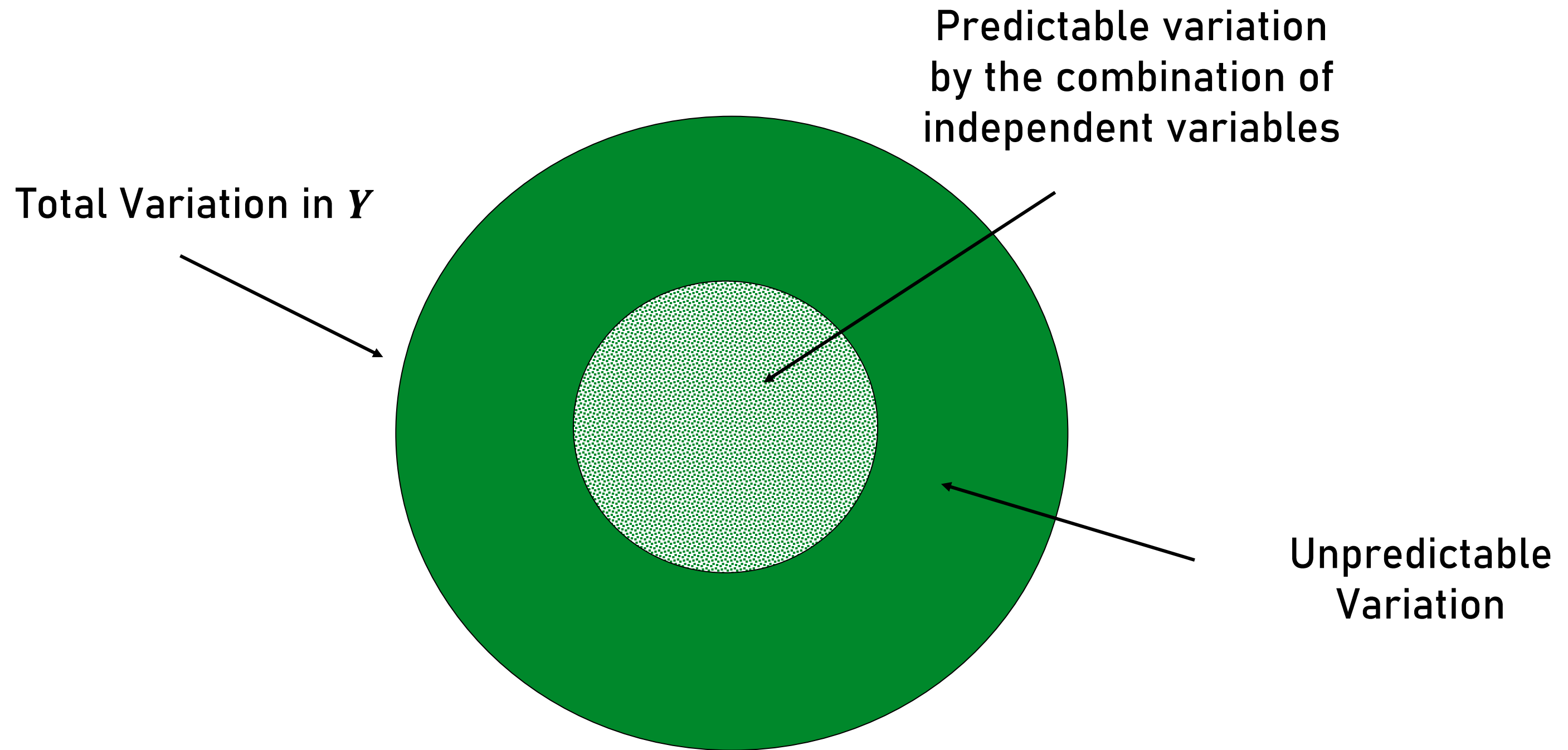
Multiple R-squared: 0.61, Adjusted R-squared: 0.607

F-statistic: 171 on 3 and 327 DF, p-value: <0.00000000000000002

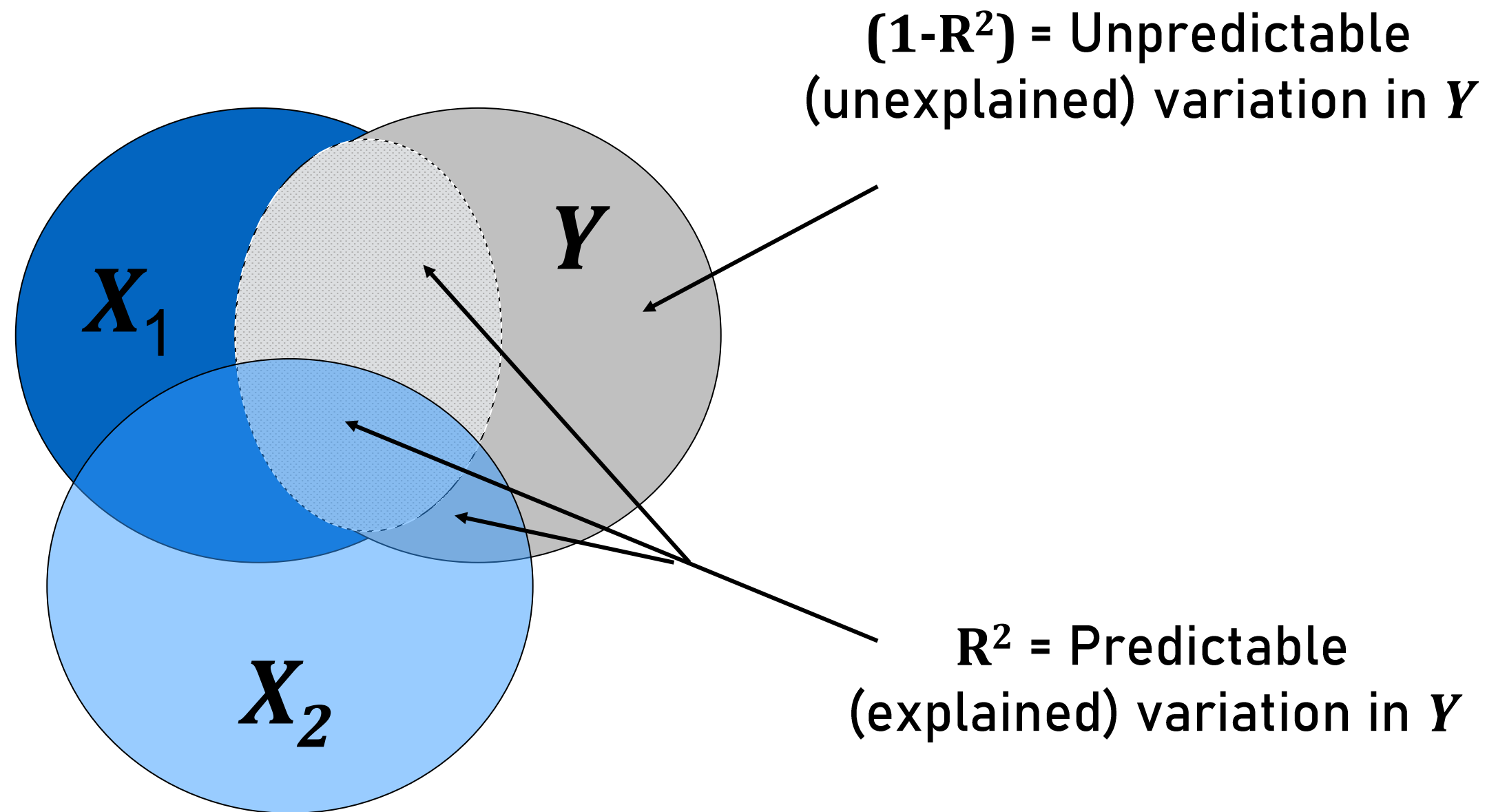
Coefficient of determination R^2

- The square of the multiple correlation coefficient (R)
- Report (adjusted) R^2 instead of R
- Indicates the % of variance in the dependent variable explained by the independent variables together.
- The adjusted R^2 takes into account both the number of variables in the model and the sample size.

Explaining Variation: How much?



Proportion of Predictable and Unpredictable Variation



R^2 can be inflated by adding lots of predictors into the model even if most of these predictors are frivolous

Interpretation of R^2

- $R^2 = 0$: the model explains none of the variation in Y ; $R^2 = 1$: it explains all of it.
- There are no universal cut-offs: what counts as a good R^2 depends on the field and the data.
- Individual-level survey data (e.g. FRS): R^2 is often low, below 0.3.
- Area-level data (e.g. Census districts): R^2 is often much higher, 0.5 or more.
- A low R^2 does not make significant coefficients meaningless: other factors also matter.

Building Regression Models

- **Simultaneous:** all independent variables entered together
- **Stepwise:** independent variables entered according to some order (e.g. by size or correlation with dependent variable).